

Supplementary Material

Bias from small-count suppression in county-level cancer disparity estimates: a calibrated simulation study

Supplementary Table S1. Full simulation grid (108 scenarios)

Population-weighted complete-case rate ratio versus known truth. beta = log rate ratio per SD of deprivation; corr = deprivation-county-size correlation. 1,000 replicates/cell.

base_rate	beta_logRR_perSD	corr_pop_dep	threshold	mean_suppressed	RR_true	RR_cc	RR_relbias
5	0	0	10	0.571	1	1	-0.0002
5	0	0	16	0.692	1	1	-0
5	0	0	25	0.79	1	1	-0.0001
5	0	-0.3	10	0.571	1	1.019	0.0193
5	0	-0.3	16	0.692	1	1.015	0.0155
5	0	-0.3	25	0.79	1	1.013	0.0127
5	0	-0.6	10	0.571	1	1.063	0.0634
5	0	-0.6	16	0.692	1	1.05	0.0501
5	0	-0.6	25	0.79	1	1.041	0.0407
5	0.262	0	10	0.57	2.085	2.063	-0.0106
5	0.262	0	16	0.689	2.088	2.069	-0.0093
5	0.262	0	25	0.786	2.086	2.068	-0.0088
5	0.262	-0.3	10	0.574	2.091	2.107	0.0075
5	0.262	-0.3	16	0.699	2.093	2.108	0.0073
5	0.262	-0.3	25	0.799	2.093	2.103	0.0049
5	0.262	-0.6	10	0.578	2.126	2.204	0.0366
5	0.262	-0.6	16	0.71	2.125	2.185	0.0281
5	0.262	-0.6	25	0.813	2.125	2.168	0.0201
5	0.531	0	10	0.567	4.457	4.335	-0.0275
5	0.531	0	16	0.682	4.437	4.316	-0.0275
5	0.531	0	25	0.777	4.454	4.337	-0.0266
5	0.531	-0.3	10	0.574	4.44	4.423	-0.0039
5	0.531	-0.3	16	0.7	4.43	4.417	-0.0032
5	0.531	-0.3	25	0.8	4.436	4.425	-0.003
5	0.531	-0.6	10	0.584	4.55	4.656	0.0231
5	0.531	-0.6	16	0.724	4.536	4.618	0.0176
5	0.531	-0.6	25	0.831	4.542	4.62	0.0165
15	0	0	10	0.296	1	1	-0.0002
15	0	0	16	0.413	1	1	0.0001
15	0	0	25	0.532	1	1	-0.0004
15	0	-0.3	10	0.296	1	1.007	0.0072
15	0	-0.3	16	0.413	1	1.007	0.0067
15	0	-0.3	25	0.532	1	1.006	0.0063
15	0	-0.6	10	0.296	1	1.026	0.0259
15	0	-0.6	16	0.413	1	1.025	0.0248
15	0	-0.6	25	0.532	1	1.021	0.0213
15	0.262	0	10	0.298	2.087	2.078	-0.0043
15	0.262	0	16	0.414	2.087	2.078	-0.0044
15	0.262	0	25	0.532	2.083	2.074	-0.0045
15	0.262	-0.3	10	0.288	2.091	2.096	0.0025
15	0.262	-0.3	16	0.409	2.093	2.097	0.0023
15	0.262	-0.3	25	0.533	2.092	2.096	0.0021
15	0.262	-0.6	10	0.278	2.128	2.155	0.013
15	0.262	-0.6	16	0.404	2.125	2.148	0.0109
15	0.262	-0.6	25	0.536	2.127	2.143	0.0075
15	0.531	0	10	0.306	4.453	4.399	-0.0122
15	0.531	0	16	0.418	4.446	4.384	-0.0141

base_rate	beta_logRR_perSD	corr_pop_dep	threshold	mean_suppressed	RR_true	RR_cc	RR_relbias
15	0.531	0	25	0.53	4.436	4.365	-0.0162
15	0.531	-0.3	10	0.287	4.435	4.429	-0.0013
15	0.531	-0.3	16	0.409	4.443	4.438	-0.0013
15	0.531	-0.3	25	0.534	4.428	4.418	-0.0025
15	0.531	-0.6	10	0.262	4.54	4.571	0.0067
15	0.531	-0.6	16	0.397	4.544	4.57	0.0057
15	0.531	-0.6	25	0.538	4.551	4.576	0.0052
50	0	0	10	0.093	1	1	0
50	0	0	16	0.156	1	1	0
50	0	0	25	0.237	1	1	0
50	0	-0.3	10	0.093	1	1.002	0.0015
50	0	-0.3	16	0.156	1	1.002	0.0018
50	0	-0.3	25	0.238	1	1.002	0.0021
50	0	-0.6	10	0.093	1	1.006	0.0059
50	0	-0.6	16	0.156	1	1.007	0.0069
50	0	-0.6	25	0.237	1	1.008	0.0079
50	0.262	0	10	0.096	2.084	2.082	-0.0008
50	0.262	0	16	0.159	2.087	2.084	-0.0011
50	0.262	0	25	0.241	2.086	2.083	-0.0013
50	0.262	-0.3	10	0.085	2.091	2.091	0.0004
50	0.262	-0.3	16	0.147	2.094	2.096	0.0007
50	0.262	-0.3	25	0.229	2.091	2.093	0.0006
50	0.262	-0.6	10	0.073	2.126	2.131	0.0024
50	0.262	-0.6	16	0.133	2.129	2.133	0.0018
50	0.262	-0.6	25	0.215	2.126	2.129	0.0015
50	0.531	0	10	0.106	4.46	4.448	-0.0026
50	0.531	0	16	0.17	4.44	4.421	-0.0042
50	0.531	0	25	0.25	4.445	4.42	-0.0057
50	0.531	-0.3	10	0.084	4.432	4.43	-0.0004
50	0.531	-0.3	16	0.145	4.436	4.433	-0.0005
50	0.531	-0.3	25	0.228	4.435	4.434	-0.0003
50	0.531	-0.6	10	0.058	4.555	4.56	0.001
50	0.531	-0.6	16	0.115	4.544	4.55	0.0013
50	0.531	-0.6	25	0.197	4.549	4.553	0.0009
150	0	0	10	0.021	1	1	-0
150	0	0	16	0.042	1	1	0.0001
150	0	0	25	0.075	1	1	-0.0002
150	0	-0.3	10	0.021	1	1	0
150	0	-0.3	16	0.042	1	1	0.0003
150	0	-0.3	25	0.075	1	1.001	0.0005
150	0	-0.6	10	0.021	1	1.001	0.0007
150	0	-0.6	16	0.041	1	1.001	0.0013
150	0	-0.6	25	0.075	1	1.002	0.0019
150	0.262	0	10	0.022	2.086	2.085	-0.0002
150	0.262	0	16	0.044	2.082	2.081	-0.0002
150	0.262	0	25	0.078	2.087	2.087	-0.0004
150	0.262	-0.3	10	0.017	2.092	2.093	0.0001
150	0.262	-0.3	16	0.036	2.092	2.092	0.0003
150	0.262	-0.3	25	0.067	2.094	2.094	0.0001
150	0.262	-0.6	10	0.013	2.128	2.128	0.0002
150	0.262	-0.6	16	0.028	2.126	2.127	0.0002
150	0.262	-0.6	25	0.056	2.126	2.126	0.0001
150	0.531	0	10	0.027	4.448	4.446	-0.0004
150	0.531	0	16	0.05	4.457	4.454	-0.0008
150	0.531	0	25	0.087	4.448	4.442	-0.0013
150	0.531	-0.3	10	0.016	4.429	4.428	-0.0001
150	0.531	-0.3	16	0.035	4.434	4.434	-0.0001

base_rate	beta_logRR_perSD	corr_pop_dep	threshold	mean_suppressed	RR_true	RR_cc	RR_relbias
150	0.531	-0.3	25	0.066	4.436	4.436	-0
150	0.531	-0.6	10	0.008	4.554	4.555	0.0002
150	0.531	-0.6	16	0.02	4.552	4.552	0.0001
150	0.531	-0.6	25	0.043	4.553	4.554	0.0001

Supplementary Table S2. Calibrated sensitivity (corr = -0.30)

Population-weighted vs unweighted; Poisson (overdisp 0) and overdispersed (0.5).

base_rate	corr_pop_dep	threshold	overdisp	mean_suppressed	RR_true	RR_cc_wt	RR_cc_unwt	relbias_wt	relbias_unwt
5	-0.3	16	0	0.7	4.425	4.413	4.424	-0.003	-0
5	-0.3	16	0.5	0.731	4.437	4.381	4.351	-0.013	-0.019
15	-0.3	16	0	0.409	4.433	4.425	4.423	-0.002	-0.002
15	-0.3	16	0.5	0.479	4.439	4.406	4.336	-0.007	-0.023
50	-0.3	16	0	0.145	4.43	4.429	4.434	-0	0.001
50	-0.3	16	0.5	0.22	4.444	4.458	4.371	0.003	-0.016
150	-0.3	16	0	0.035	4.442	4.442	4.439	0	-0.001
150	-0.3	16	0.5	0.079	4.437	4.477	4.401	0.009	-0.008

Supplementary Table S3. STROBE checklist (cross-sectional)

STROBE item	Recommendation	Location
1 Title/abstract	Design in title; structured abstract	Title; Abstract
2 Background	Scientific background, rationale	Introduction
3 Objectives	Stated objective/aim	Introduction (aim)
4 Study design	Key design elements	Methods 1
5 Setting	Setting, locations, dates	Methods 1,4
6 Participants	Units, eligibility	Methods 2
7 Variables	Exposure, outcome, covariates	Methods 3-4
8 Data sources/measurement	Sources, measurement	Methods 1,3,4
9 Bias	Sources of bias addressed	Methods 6; Discussion
10 Study size	How arrived at	Methods 2,7
11 Quantitative variables	Handling, groupings	Methods 3,5
12 Statistical methods	Methods, subgroups, sensitivity	Methods 5-9
13 Participants (results)	Numbers at each stage	Results (sample)
14 Descriptive data	Characteristics	Table 1
15 Outcome data	Outcomes/events	Tables 2-3
16 Main results	Estimates + CIs	Tables 2; Figs 1,3
17 Other analyses	Subgroups/sensitivity/sim	Methods 7-9; Fig 3
18 Key results	Summary re objectives	Discussion 1
19 Limitations	Limitations, bias direction	Discussion 5
20 Interpretation	Cautious overall interpretation	Discussion 2-3,7
21 Generalisability	External validity	Discussion 2,6
22 Funding	Funding role	Statements

Simulation reporting follows ADEMP (Morris et al., 2019): aims, data-generating mechanisms, estimands, methods, and performance measures (bias, percent bias, Monte Carlo SE) are specified in Methods section 7.